Correlation and Regression

(ii)

x	y	x^2	y^2	xy
56	147	3136	21609	8232
42	125	1764	15625	5250
36	118	1296	13924	4248
47	128	2209	16384	6016
49	125	2401	15625	6125
42	140	1764	19600	5880
72	155	5184	24025	11160
63	160	3969	25600	10080
55	149	3025	22201	8195
60	150	3600	22500	9000
$\Sigma x = 522$	$\Sigma y = 1397$	$\Sigma x^2 = 28348$	$\Sigma y^2 = 197093$	$\Sigma xy = 74186$

(i) Correlation coefficient, $r_{xy} = \dfrac{n\Sigma xy - \Sigma x \Sigma y}{\sqrt{\{n\Sigma x^2 - (\Sigma x)^2\}\{n\Sigma y^2 - (\Sigma y)^2\}}}$

$$= \frac{10 \times 74186 - (522 \times 1397)}{\sqrt{\{(10 \times 28348) - (522)^2\}\{(10 \times 197093 - (1397)^2\}}}$$

$= 0.866$ Ans.

(iii) The regression equation of y on x is

$$y = a + bx \quad \text{------------ (i)}$$

$$a = \bar{y} - b\bar{x}$$

$$= \frac{\Sigma y}{n} - b \cdot \frac{\Sigma x}{n}$$

$$= \frac{1397}{10} - b \cdot \frac{522}{10} \quad \text{------------ (ii)}$$

We know,

$$b = \frac{n\Sigma xy - \Sigma x \Sigma y}{n\Sigma x^2 - (\Sigma x)^2}$$

$$= \frac{10 \times 74186 - (522 \times 1397)}{10 \times 28348 - (522)^2}$$

$$= \frac{741860 - 729234}{283480 - 272484}$$

$$= \frac{12626}{10996} = 1.148$$

From, (ii) we get,

$$a = \frac{1397}{10} - 1.148 \cdot \frac{522}{10}$$

$$= 139.7 - 59.9256 = 79.77$$

CamScanner

$\therefore$ The regression line of y on x;

$$\hat{y} = -79.77 + 1.148x \text{ Ans.}$$

(iv)

Now, Let $x = 50$

So, $\hat{y} = -79.77 + (1.148 \times 50)$

$= -22.37$ Ans.

(v) The regression equation of x on y is,

$$\hat{x} = a + by \quad \text{——— (i)}$$

$a = \bar{x} - b\bar{y}$

$= \frac{\Sigma x}{n} - b.\frac{\Sigma y}{n}$

$= \frac{522}{10} - b.\frac{1397}{10} \quad \text{——— (ii)}$

we know,

$$b = \frac{n\Sigma xy - \Sigma x \Sigma y}{n\Sigma y^2 - (\Sigma y)^2}$$

$$= \frac{10 \times 74186 - 522 \times 1397}{10 \times 197093 - (1397)^2}$$

$$= \frac{12626}{19321} = 0.65$$

From equation (iii),

$$a = \frac{522}{10} - 0.65 \frac{1397}{10}$$

$$= 52.2 - 90.805$$

$$= -38.61$$

From equation (1),

$$\hat{x} = -38.61 + 0.65y$$ Ans.

Fundamentals of Probability

(1) Given that, $P[A] = 0.6$

$P[B] = 0.8$

$P[AB] = 0.50$

(i) $P[\bar{A}]$.

We know that, $P[A] + P[\bar{A}] = 1$

$\therefore P[\bar{A}] = 1 - P[A] = 1 - 0.6 = 0.4$

(ii) $P[A \cup B]$

We know that, $P[A \cup B] = P[A] + P[B] - P[AB]$

$= 0.6 + 0.8 - 0.50$

$= 0.9$

(iii) $P[A|B]$

We know,

$$P[A|B] = \frac{P[AB]}{P[B]}$$

$$= \frac{0.50}{0.8} = 0.625$$

(iv) $P[B|A]$

We know, $P[B|A] = \frac{P[AB]}{P[A]} = \frac{0.50}{0.6}$

$$= 0.833$$

(v) $P[A\bar{B}]$

$$P[A\bar{B}] = P[A \cap \bar{B}] = P[A] - P[AB]$$

$$= 0.6 - 0.50 = 0.10$$

(vi) $P[\bar{A}B]$

$$P[\bar{A}B] = P[\bar{A} \cap B] = P[B] - P[AB] = 0.8 - 0.50$$

$$= 0.30$$

(vii) $P[\bar{A}\bar{B}]$

By De Morgans law,

$P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B})$.

We know,

$$P(A \cup B) + P[\overline{A \cup B}] = 1$$

$$\text{or,} \quad P[\overline{A \cup B}] = 1 - P[A \cup B]$$
$$= 1 - 0.9$$
$$= 0.1$$

(viii) $P[\overline{A \cup B}]$

From (vii) $= 0.1$

(ix) events A and B will be independent if

$$P[A]P[B] = P[AB]$$

Now, $P[A]P[B] = 0.6 \times 0.8$
$$= 0.48 \neq 0.50$$

As $P[A]P[B] \neq P[AB]$

Therefore, events A and B are not independents.

(x) A and B are mutually exclusive if

$$P[AB] = 0$$

Here, $P[AB) = 0.50$

So, A and B are non mutually exclusive.

② A and B are mutually exclusive.

$\therefore$ $P[AB] = 0$

Given, $P[A] = 0.35$

$P[B] = 0.15$

(i) $P[A \cup B]$

We know,

$$P[A \cup B] = P[A] + P[B] - P[AB]$$

$$= 0.35 + 0.15 - 0$$

$$= 0.5$$

(ii) $P[\overline{A}]$

We know,

$$P[A] + P[\overline{A}] = 1$$

or, $P[\overline{A}] = 1 - P[A]$

or, $P[\overline{A}] = 1 - 0.35$

$= 0.65$

(iii) $P[A \cap B] = 0$

(iv) $P[\overline{A} \cup \overline{B}] = P[\overline{A \cap B}]$

So, $P[\overline{A \cap B}] = 1 - P[P \cap B] = 1 - 0 = 1$

(3) Given that,

$$P(A) = \frac{1}{8}$$

$$P(A|B) = \frac{1}{4}$$

$$P(B) = \frac{1}{6}$$

(i) A and B are independent if $P(A|B) = P(A)$

Here, $P(A|B) = \frac{1}{4}$

$P(A) = \frac{1}{8}$

So, $P(A/B) \neq P(A)$

Therefore, A and B are not independent.

(ii) A and B are mutually exclusive if $P[AB] = 0$.

We know,

$$P(A|B) = \frac{P[AB]}{P[B]}$$

or, $P[AB] = P[A|B]P[B]$

$$= \frac{1}{4} \cdot \frac{1}{6}$$

$$= \frac{1}{24} \neq 0$$

So, A and B are ~~not~~ non mutually exclusive.

CamScanner

(iii) From (ii) we get,

$$P(A \cap B) = P[AB] = \frac{1}{24}$$

We know,

$$P[A \cup B] = P[A] + P[B] - P[A \cap B]$$

$$= \frac{1}{8} + \frac{1}{6} - \frac{1}{24}$$

$$= \frac{1}{4}$$

Random Variable and Mathematical Expectation

(1)

(ii) $E[X]$

We know, $E[X] = \int_{-\infty}^{\infty} x f(x)\, dx$

Now, $E[X] = \int_0^2 x \cdot \frac{x}{2} \cdot dx$

$= \frac{1}{2}\int_0^2 x^2 \cdot dx = \frac{1}{2}\left[\frac{x^3}{3}\right]_0^2$

$= \frac{1}{2}\left(\frac{8}{3} - 0\right)$

$= \frac{4}{3}$

(iii) $V[X]$.

We know, $V[X] = E[X^2] - (E[X])^2$

$= \int_0^2 x^2 f(x)\, dx - \left(\frac{4}{3}\right)^2$

$= \int_0^2 x^2 \cdot \frac{x}{2} dx - \left(\frac{4}{3}\right)^2$

CamScanner

$$= \frac{1}{2}\int_0^2 x^3 dx - \left(\frac{4}{3}\right)^2$$

$$= \frac{1}{2}\left[\frac{x^4}{4}\right]_0^2 - \left(\frac{4}{3}\right)^2$$

$$= \frac{1}{2}\left(\frac{16}{4}\right) - \left(\frac{4}{3}\right)^2$$

$$= \frac{8}{4} - \frac{16}{9}$$

$$= 2 - \frac{16}{9} = \frac{2}{9}$$ Ans

(iv) $P[0.75 \le x \le 1.5]$

$$\int_{0.75}^{1.5} \frac{x}{2} dx = \frac{1}{2}\left[\frac{x^2}{2}\right]_{0.75}^{1.5}$$

$$= \frac{1}{2}\left\{\frac{(1.5)^2}{2} - \frac{(0.75)^2}{2}\right\}$$

$$= \frac{1}{2}\left(\frac{2.25}{2} - \frac{0.5625}{2}\right)$$

$$= \frac{1}{2} \cdot \frac{1.6875}{2}$$

$$= \frac{1.6875}{4} = 0.42$$ Ans

(i) $P[x \leq 1.25]$

$$\int_0^{1.25} f(x)\,dx = \int_0^{1.25} \frac{x}{2}\,dx = \frac{1}{2}\left[\frac{x^2}{2}\right]_0^{1.25}$$

$$= \frac{1}{2}\left\{\frac{(1.25)^2}{2} - 0\right\} = \frac{1}{2} \times \frac{25}{32} = 0.39.$$

(v) $E[5x+7]$

We know, $E[ax+b] = a\,E[x] + b$

Now, $E[5x+7] = 5\,E[x] + 7$

$$= 5 \times \frac{4}{3} + 7$$

$$= \frac{20}{3} + 7 = 13.67$$

(vi) $V[5x+7]$

We know; $V[ax+b] = a^2 V(x)$

From (iii),

$V[x] = \frac{2}{9}$

$\therefore V[5x+7] = (5)^2 \cdot \frac{2}{9} = 25 \times \frac{2}{9} = 5.56$

Problem:

A continuous random variable x has the following probability desing function:

$$f(x) = kx^2 \; ; \; 0 \leq x \leq 1$$

① Determine the value of k

② $P(X < 0.65)$

③ $P(x > 0.30)$

④ $P(0.25 < x < 0.75)$

⑤ $E[x]$ or $E[7x+9]$

⑥ $V(x)$ or $V(7x+9)$

⑦ $SD(x)$

Solution:

① We know, $\int_{-\infty}^{\infty} f(x)\,dx = 1$

or, $\int_0^1 kx^2\,dx = 1$

or, $k\int_0^1 x^2\,dx = 1$

or, $k\left[\frac{x^3}{3}\right]_0^1 = 1$

or, $\frac{k}{3}\left[x^3\right]_0^1 = 1$

or, $\frac{k}{3}(1^3 - 0^3) = 1$

or, $\frac{k}{3} \cdot 1 = 1$

or, $k = 3$

Therefore, $f(x) = 3x^2 \; ;$

$0 \leq x \leq 1$

② $P(x < 0.65)$

$$= \int_0^{0.65} 3x^2\,dx$$

$$= 3\int_0^{0.65} x^2\,dx \qquad = 3\left[\frac{x^3}{3}\right]_0^{0.65}$$

$$\cancel{= 3\int_0^{0.65}} \qquad = \cancel{3}\left[x^3\right]_0^{0.65}$$

$$= (0.65)^3 - (0)^3$$

$$= 0.27$$

③ $P(x > 0.30)$

$$= \int_{0.30}^{1} 3x^2\,dx$$

$$= 3\int_{0.30}^{1} x^2\,dx$$

$$= 3\left[\frac{x^3}{3}\right]_{0.30}^{1}$$

$$= \left[x^3\right]_{0.30}^{1}$$

$$= (1)^3 - (0.30)^3$$

$$= 1 - 0.027 = 0.973$$

④ $P(0.25 < x < 0.75)$

$$= \int_{0.25}^{0.75} 3x^2\,dx = 3\int_{0.25}^{0.75} x^2\,dx = 3\left[\frac{x^3}{3}\right]_{0.25}^{0.75}$$

$$= (0.75)^3 - (0.25)^3 = 0.389 - 0.156 = 0.233$$

⑤ $E[x] = \int_{-\infty}^{\infty} x f(x)\,dx$

$$= \int_0^1 x\, 3x^2\,dx$$

$$= 3\int_0^1 x^3\,dx = 3\int_0^1 x^3\,dx = 3\left[\frac{x^4}{4}\right]_0^1$$

$$= \frac{3}{4}\left((1)^4 - (0)^4\right) = \frac{3}{4}$$

and, $E[7x+9] = \cancel{7E[x+9]} = 7E[x]+9$

$$= 7\cdot\frac{3}{4} + 9 = 14.25$$

CamScanner

(6) $V(x) = E[x^2] - [E[x]]^2$

$= \int_0^1 x^2 3x^2 dx - \left(\frac{3}{4}\right)^2$

$= 3\int_0^1 x^4 dx - \left(\frac{3}{4}\right)^2$

$= 3\left[\frac{x^5}{5}\right]_0^1 - \left(\frac{3}{4}\right)^2$

$= \frac{3}{5}(1^5 - 0) - \left(\frac{3}{4}\right)^2$

$= \frac{3}{5} - \frac{9}{16}$

$= \frac{48 - 45}{80}$

$= \frac{3}{80}$

and, $V(7x+9) = (7)^2 \cdot V(x) + 0$

$= 49 \cdot \frac{3}{80} + 0$

$= 1.8375$

7 $SD(x) = \sqrt{V(x)}$

$= \sqrt{\frac{3}{80}}$ (Ans.)

(3) Given that,

$$f(x) = k(2x - x^2) \; ; \; 0 < x < 2$$

(i) We know,

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

or, $$\int_0^2 k(2x - x^2)\,dx = 1$$

or, $$k\left[2 \cdot \frac{x^2}{2} - \frac{x^3}{3}\right]_0^2 = 1$$

or, $$k\left[x^2 - \frac{x^3}{3}\right]_0^2 = 1$$

or, $$k\left\{(2)^2 - \frac{(2)^3}{3} - 0\right\} = 1$$

or, $$k\left(4 - \frac{8}{3}\right) = 1$$

or, $$k \cdot \frac{4}{3} = 1$$

or, $$k = \frac{3}{4}$$

(ii) $P(x>1)$

from (i) $f(x) = \frac{3}{4}(2x - x^2)$

$$\int_1^2 \frac{3}{4}(2x - x^2)\,dx = \frac{3}{4}\int_1^2 (2x - x^2)\,dx$$

$$= \frac{3}{4}\left[2 \cdot \frac{x^2}{2} - \frac{x^3}{3}\right]_1^2$$

$$= \frac{3}{4}\left[x^2 - \frac{x^3}{3}\right]_1^2$$

$$= \frac{3}{4}\left\{\left(4 - \frac{8}{3}\right) - \left(1 - \frac{1}{3}\right)\right\}$$

$$= \frac{3}{4}\left(\frac{4}{3} - \frac{2}{3}\right)$$

$$= \frac{3}{4} \times \frac{2}{3}$$

$$= \frac{1}{2}$$

(iii) $P(1.5 < x < 2.25)$

$$\int_{1.5}^{2.25} \frac{3}{4}(2x - x^2)\,dx = \frac{3}{4}\left[2\cdot\frac{x^2}{2} - \frac{x^3}{3}\right]_{1.5}^{2.25}$$

$$= \frac{3}{4}\left[\left\{(2.25)^2 - \frac{(2.25)^3}{3}\right\} - \left\{(1.5)^2 - \frac{(1.5)^3}{3}\right\}\right]$$

$$= \frac{3}{4}\left(\frac{81}{64} - \frac{9}{8}\right)$$

$$= 0.105$$

(iv) $E[3x+6]$

we know,

$$E(3x+6) = 3E[X] + 6$$

$$= 3\cdot\int_0^2 x\, f(x)\,dx + 6$$

$$= 3\int_0^2 x\cdot\frac{3}{4}(2x - x^2)\,dx + 6$$

$$= 3\cdot\frac{3}{4}\left[2\cdot\frac{x^2}{2} - \frac{x^3}{3}\right]_0^2 + 6$$

$$= \frac{9}{4}\left(4 - \frac{8}{3}\right) + 6$$

$$= \frac{9}{4} \cdot \frac{4}{3} + 6$$

$$= 3 + 6$$

$$= 9$$

(v) $V[4x+9]$

We know, $V[4x+9] = \cancel{2}\cdot(4)^2 V[X]$ ——— (i)

$$V(X) = E[X^2] - (E[X])^2$$

$$= \int_0^2 x^2 f(x)\,dx - (1)^2$$

$$= \int_0^2 x^2 \cdot \frac{3}{4}(2x - x^2)\,dx - 1$$

$$= \frac{3}{4}\int_0^2 (2x^3 - x^4)\,dx - 1$$

$$= \frac{3}{4}\left[2\cdot\frac{x^4}{4} - \frac{x^5}{5}\right]_0^2 - 1$$

$$= \frac{3}{4}\cdot\frac{8}{5} - 1$$

$$= \frac{1}{5}$$

Putting the value $V[X]$ in equation (i) we get,

$$V[4x+9] = 16 \times \frac{1}{5}$$

$$= \frac{16}{5}$$

(4) Given that,

$$f(x) = k(x-1) \; ; \; 2 \leq x \leq 6$$

(i) We know,

$$\int_{-\infty}^{\infty} f(x).dx = 1$$

$$\text{or, } \int_{2}^{6} k(x-1)dx = 1$$

$$\text{or, } k\int_{2}^{6} (x-1)dx = 1$$

$$\text{or, } k\left[\frac{x^2}{2} - x\right]_2^6 = 1$$

$$\text{or, } k\left\{\left(\frac{36}{2} - 6\right) - \left(\frac{4}{2} - 2\right)\right\} = 1$$

or, $K(12-0) = 1$

or, $K = \frac{1}{12}$

$\therefore f(x) = \frac{1}{12}(x-1)$

(ii) $P(x>3)$

$$\therefore \int_3^6 \frac{1}{12}(x-1)\,dx = 12\left[\frac{x^2}{2} - x\right]_3^6$$

$$= \frac{1}{12}\left\{\left(\frac{36}{2} - 6\right) - \left(\frac{9}{2} - 3\right)\right\}$$

$$= \frac{1}{12}\left(12 - \frac{3}{2}\right) = \frac{1}{12} \cdot \frac{21}{2}$$

$$= \frac{7}{8} \quad \text{Ans.}$$

(iii) $P(3<x<4)$

$$\therefore \int_3^4 \frac{1}{12}(x-1)\,dx = \frac{1}{2}\int_3^4 (x-1)\,dx = \frac{1}{2}\left[\frac{x^2}{2} - x\right]_3^4$$

$$= \frac{1}{2}\left\{\left(\frac{16}{2} - 4\right) - \left(\frac{9}{2} - 3\right)\right\}$$

$$= \frac{1}{12}\left(4 - \frac{3}{2}\right) = \frac{1}{12} \cdot \frac{5}{2} = \frac{5}{24}$$

Ans.

(iv) $E[2x+7]$

We know,

$$E[2x+7] = 2E[X]+7$$

Now,

$$E[X] = \int_2^6 x \cdot \frac{1}{12}(x-1)\,dx$$

$$= \frac{1}{12}\int_2^6 (x^2 - x)\,dx$$

$$= \frac{1}{12}\left[\frac{x^3}{3} - \frac{x^2}{2}\right]_2^6$$

$$= \frac{1}{12}\left\{(72-18) - \left(\frac{8}{3} - 2\right)\right\}$$

$$= \frac{1}{12}\left(54 - \frac{2}{3}\right)$$

$$= \frac{40}{9} \text{ Ans.}$$

Now,

$$E[2x+7] = 2 \cdot \frac{40}{9} + 7$$

$$= \frac{80}{9} + 7$$

$$= \frac{143}{9} \text{ Ans.}$$

CamScanner

(v) $V[9x+7]$

We know, $V[9x+7] = (9)^2 V[x] = 81 V[x]$

$$V[x] = E[x^2] - E[x]^2$$

$$= \int_2^6 x^2 \cdot \frac{1}{12}(x-1)\,dx - \left(\frac{40}{9}\right)^2$$

$$= \frac{1}{12}\int_2^6 (x^3 - x^2)\,dx - \left(\frac{40}{9}\right)^2$$

$$= \frac{1}{12}\left[\frac{x^4}{4} - \frac{x^3}{3}\right]_2^6 - \left(\frac{40}{9}\right)^2$$

$$= \frac{1}{12}\cdot\left\{(324-72) - \left(4 - \frac{8}{3}\right)\right\} - \left(\frac{40}{9}\right)^2$$

$$= \frac{1}{12}\left(252 - \frac{4}{3}\right) - \left(\frac{40}{9}\right)^2$$

$$= \frac{187}{9} - \frac{1600}{81}$$

$$= \frac{83}{81}$$

Now, $V[9x+7] = 81 \cdot \frac{83}{81} = 83$ Ans

(5) Given that,

$$f(x) = K(x+1);\quad 2 \le x \le 5$$

(i) We know,

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

or, $\int_{2}^{5} K(x+1)\,dx = 1$

or, $K\int_{2}^{5} (x+1)\,dx = 1$

or, $K\left[\frac{x^2}{2}+x\right]_2^5 = K\left\{\left(\frac{25}{2}+5\right)-\left(2+2\right)\right\} = 1$

or, $K \cdot \frac{27}{2} = 1$

$\therefore K = \frac{2}{27}$

(ii) $P(X>3)$

$\therefore \int_3^5 \frac{2}{27}(x+1)dx = \frac{2}{27}\int_3^5 (x+1)\,dx$

$= \frac{2}{27}\left[\frac{x^2}{2}+x\right]_3^5 = \frac{2}{27}\cdot\left(\frac{35}{2}-\frac{15}{2}\right) = \frac{20}{27}$ Ans

(iii) $P(x=4)$

(iv) $P(3<X<4)$

$\therefore \int_3^4 \frac{2}{27}(x+1)dx = \frac{2}{27}\left[\frac{x^2}{2}+x\right]_3^4 = \frac{2}{27}\left[12-\frac{15}{2}\right]$

$= \frac{1}{3}$ (Ans)

(v) $E[x+7]$

We know, $E[x+7] = E[x]+7$

$E[x] = \int_{-\infty}^{\infty} x f(x)\cdot dx = \int_2^5 x \times \frac{2}{27}(x+1)\,dx$

$= \frac{2}{27}\int_2^5 (x^2+x)dx = \frac{2}{27}\cdot \cancel{\frac{29}{2}}\left[\frac{x^3}{3}+\frac{x^2}{2}\right]_2^5$

$= \frac{2}{27}\left(\frac{325}{6} \cancel{+} - \frac{14}{3}\right) = \cancel{\frac{2}{27}}\ \frac{11}{3}$

$$= \frac{3}{12}\{(16-12)-0\}$$

$$= \frac{3}{12} \times 4 = 1 \text{ inches.}$$

Variance,

$$V[x] = E[x^2] - (E(x))^2$$

$$= \frac{3}{12}\int_0^2 x^2(6x^2-3x^3)\,dx - (1)^2$$

$$= \frac{3}{12}\int_0^2 (6x^4-3x^5)\,dx - 1$$

$$= \frac{3}{12}\left[\frac{6x^5}{5} - \frac{x^6}{2}\right]_0^2 - 1$$

$$= \frac{3}{12}\left(\frac{192}{5} - 32\right) - 1$$

$$= \frac{8}{5} - 1$$

$$= \frac{3}{5}$$

$$= 0.6 \text{ inches (Ans)}$$

PROBABILITY DISTRIBUTION

(1) Given, $n = 5$

$p = q = \frac{1}{2}$

Let 'x' be the number of boy.

(i) $x = 0$

$P(x=0) = {}^5C_0\, p^0 q^{n-0} = {}^5C_0\, p^0 q^5$

$= {}^5C_0 \left(\frac{1}{2}\right)^0 \cdot \left(\frac{1}{2}\right)^5 = 1 \cdot 1 \cdot \frac{1}{32} = \frac{1}{32}$ Ans

(ii) $x \leq 2$

$P(x \leq 2) = f(0) + f(1) + f(2)$

$= {}^5C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^5 + {}^5C_1 \left(\frac{1}{2}\right)\left(\frac{1}{2}\right)^4 + {}^5C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^3$

$= 1 \cdot 1 \cdot \frac{1}{32} + 5 \cdot \frac{1}{2} \cdot \frac{1}{16} + 10 \cdot \frac{1}{4} \cdot \frac{1}{8}$

$= \frac{1}{32} + \frac{5}{32} + \frac{10}{32}$

$= \frac{16}{32}$

$= \frac{1}{2}$ Ans.

CamScanner

(ii) $x \geq 3$

$$P(x \geq 3) = f(3) + f(4) + f(5)$$

$$= {}^5C_3\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^2 + {}^5C_4\left(\frac{1}{2}\right)^4\left(\frac{1}{2}\right)^1 + {}^5C_5\left(\frac{1}{2}\right)^5\left(\frac{1}{2}\right)^0$$

$$= 10 \cdot \frac{1}{8} \cdot \frac{1}{4} + 5 \cdot \frac{1}{16} \cdot \frac{1}{2} + 1 \cdot \frac{1}{32} \cdot 1$$

$$= \frac{10}{32} + \frac{15}{32} + \frac{1}{32}$$

$$= \frac{16}{32} = \frac{1}{2}$$ Ans.

(2) Given,

$n = 5$

$p = q = \frac{1}{2}$ [As it is a fair coin]

~~so~~ Let x be the number of head

C.W

③ Given,

number of trials, $n = 10$

$p = q = \frac{1}{2}$

Let, x be the number of heads.

So, $f(x) = {}^{10}C_x \, p^x q^{10-x} \quad [x = 0, 1, 2, \ldots \ldots, 10]$

(i) At least 7 heads

$\cancel{x \geq 7}$; $x \geq 7$

$p(x \geq 7) = f(7) + f(8) + f(9) + f(10)$

$= {}^{10}C_7 p^7 q^3 + {}^{10}C_8 p^8 q^2 + {}^{10}C_9 p^9 q^1 + {}^{10}C_{10} p^{10} q^0$

$= {}^{10}C_7 \left(\frac{1}{2}\right)^7 \left(\frac{1}{2}\right)^3 + {}^{10}C_8 \left(\frac{1}{2}\right)^8 \left(\frac{1}{2}\right)^2 + {}^{10}C_9 \left(\frac{1}{2}\right)^9 \left(\frac{1}{2}\right)^1 + {}^{10}C_{10} \left(\frac{1}{2}\right)^{10} \cdot 1$

$= 120 \cdot \frac{1}{128} \cdot \frac{1}{8} + 45 \cdot \frac{1}{256} \cdot \frac{1}{4} + 10 \cdot \frac{1}{512} \cdot \frac{1}{2} + 1 \cdot \frac{1}{1024} \cdot 1$

$= \frac{120}{1024} + \frac{45}{1024} + \frac{10}{1024} + \frac{1}{1024}$

$= \frac{176}{1024} = \frac{11}{64}$ Ans.

(ii) Exactly 7 heads

$x = 7$

$P(x=7) = f(7) = {}^{10}C_7 \left(\frac{1}{2}\right)^7 \left(\frac{1}{2}\right)^3$

$= \frac{120}{1024} = \frac{15}{128}$ (Ans)

(iii) at most 7 heads

$x \le 7$

$P(x \le 7) = f(0) + f(1) + f(2) + f(3) + f(4) + f(5) + f(6) + f(7)$

$= {}^{10}C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{10} + {}^{10}C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^9 + {}^{10}C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^8$

$+ {}^{10}C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^7 + {}^{10}C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^6$

$+ {}^{10}C_5 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^5 + {}^{10}C_6 \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^4 + {}^{10}C_7 \left(\frac{1}{2}\right)^7 \left(\frac{1}{2}\right)^3$

$= 1 \cdot \frac{1}{1024} + 10 \cdot \frac{1}{2} \cdot \frac{1}{512} + 45 \cdot \frac{1}{4} \cdot \frac{1}{256} + 120 \cdot \frac{1}{8} \cdot \frac{1}{128}$

$+ 210 \cdot \frac{1}{16} \cdot \frac{1}{64} + 252 \cdot \frac{1}{32} \cdot \frac{1}{32} + 210 \cdot \frac{1}{64} \cdot \frac{1}{16}$

$+ \frac{120}{1024}$

$= \frac{1}{1024} + \frac{10}{1024} + \frac{45}{1024} + \frac{120}{1024} + \frac{210}{1024} + \frac{252}{1024} + \frac{210}{1024}$

$+ \frac{120}{1024}$

$= \frac{968}{1024} = \frac{121}{128}$ Ans.

(4)

Here,

20 out of 100 wrist watches are defective.

$$p = \frac{20}{100} = \frac{1}{5}$$

$$q = 1 - p = 1 - \frac{1}{5} = \frac{4}{5}$$

$$n = 10$$

$$P(X=x) = {}^{10}C_x\left(\frac{1}{5}\right)^x\left(\frac{4}{5}\right)^{10-x}$$ where, $x = 0, 1, \text{----}$

(i) 10 are defective

$$P(x=10) = {}^{10}C_{10}\left(\frac{1}{5}\right)^{10}\left(\frac{4}{5}\right)^0$$

$$= 1 \cdot \frac{1}{5^{10}} \cdot 1$$

$$= \frac{1}{5^{10}}$$

(ii) ~~not~~ 10 are good mean no defective

$$x = 0$$

$$P(x=0) = {}^{10}C_0\left(\frac{1}{5}\right)^0\left(\frac{4}{5}\right)^{10}$$

$$= 1 \cdot 1 \cdot \frac{4^{10}}{5^{10}}$$

$$= \frac{4^{10}}{5^{10}}$$ Ans

(iii) at least one defective watch

$$P(x \geq 1) = 1 - P(x < 1)$$

$$= 1 - P(x=0)$$

$$= 1 - {}^{10}C_0 \left(\frac{1}{5}\right)^0 \left(\frac{4}{5}\right)^{10}$$

$$= 1 - \left(\frac{4}{5}\right)^{10}$$

(iv) at most 3 defective watch

~~P(x≤3~~

$$P(x \leq 3) = P(0) + P(1) + P(2) + P(3)$$

$$= {}^{10}C_0 \left(\frac{1}{5}\right)^0 \left(\frac{4}{5}\right)^{10} + {}^{10}C_1 \left(\frac{1}{5}\right)^1 \left(\frac{4}{5}\right)^9 + {}^{10}C_2 \left(\frac{1}{5}\right)^2 \left(\frac{4}{5}\right)^8$$

$$+ {}^{10}C_3 \left(\frac{1}{5}\right)^3 \left(\frac{4}{5}\right)^7$$

$$= 1.1\left(\frac{4}{5}\right)^{10} + 10\left(\frac{1}{5}\right)^1 \left(\frac{4}{5}\right)^9 + 45\left(\frac{1}{5}\right)^2 \left(\frac{4}{5}\right)^8 + 120\left(\frac{1}{5}\right.$$

$$= 1.(0.107) + 10(0.026) + 45(0.0062) + 120(0.0016)$$

$$= 0.859 \text{ (approximately)}$$

Compute the value of Chi-square for above data.

Solution:

Computation table:

Result	Job satisfaction		Total
	Yes	NO	
Excellent	20 (O_{11})	70 (O_{12})	90
Good	45 (O_{21})	65 (O_{22})	110
Total	65	135	200

We know,

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Here,

$$E_{11} = \frac{90 \times 65}{200} = 29.25$$

$$E_{12} = \frac{90 \times 135}{200} = 60.75$$

$$E_{21} = \frac{110 \times 65}{200} = 35.75$$

$$E_{22} = \frac{110 \times 135}{200} = 74.25$$

Therefore,

$$\chi^2 = \frac{(20-29.25)^2}{29.25} + \frac{(70-60.75)^2}{60.75} + \frac{(45-35.75)^2}{35.75}$$

$$+ \frac{(65-74.25)^2}{74.25}$$

$$= 2.93 + 1.41 + 2.39 + 1.15$$

$$= 7.88$$

(Ans.)

Estimation and test of Hypothesis

(2)

Computation Table:

Treatment	~~Satisfaction~~ Remission of Symptom		Total
	No	Yes	
~~Excellent~~ Antibiotic	110 (O_{11})	190 (O_{12})	300
Homeopathic ~~Good~~	160 (O_{21})	140 (O_{22})	300
Total	270	330	600

We know, $$\chi^2 = \sum \left(\frac{O_{ij} - E_{ij}}{E_{ij}} \right)^2$$

$$E_{11} = \frac{300 \times 270}{600} = 135$$

$$E_{12} = \frac{300 \times 330}{600} = 165$$

$$E_{21} = \frac{300 \times 270}{600} = 135$$

$$E_{22} = \frac{300 \times 270}{600} = 165$$

$$\therefore \chi^2 = \left(\frac{110 - 135}{135} \right)^2 + \left(\frac{190 - 165}{165} \right)^2 + \left(\frac{160 - 135}{135} \right)^2 + \left(\frac{140 - 165}{165} \right)^2$$

$$= 0.034 + 0.023 + 0.034 + 0.023$$

$$= 0.114$$ Ans.